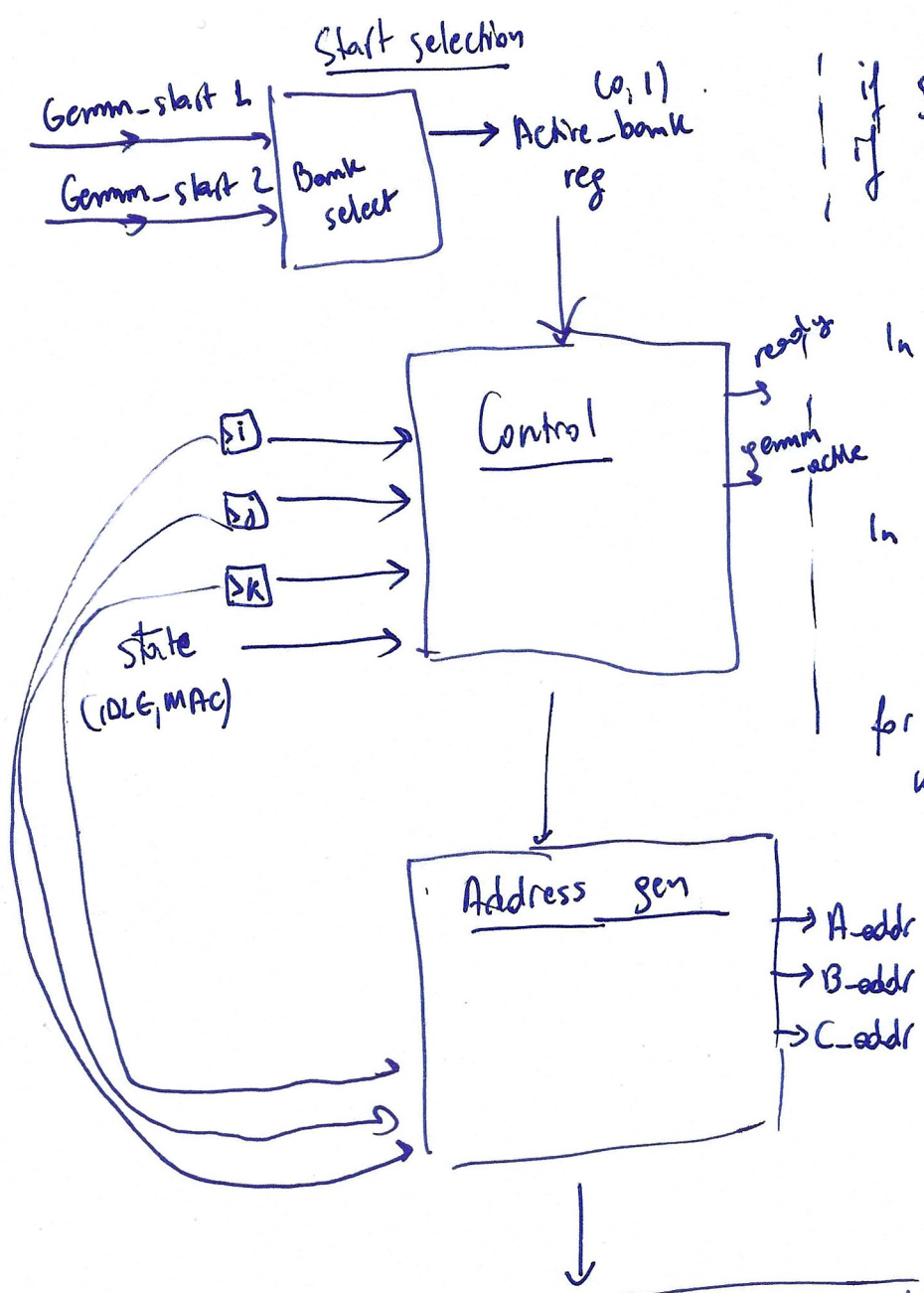


GEMM



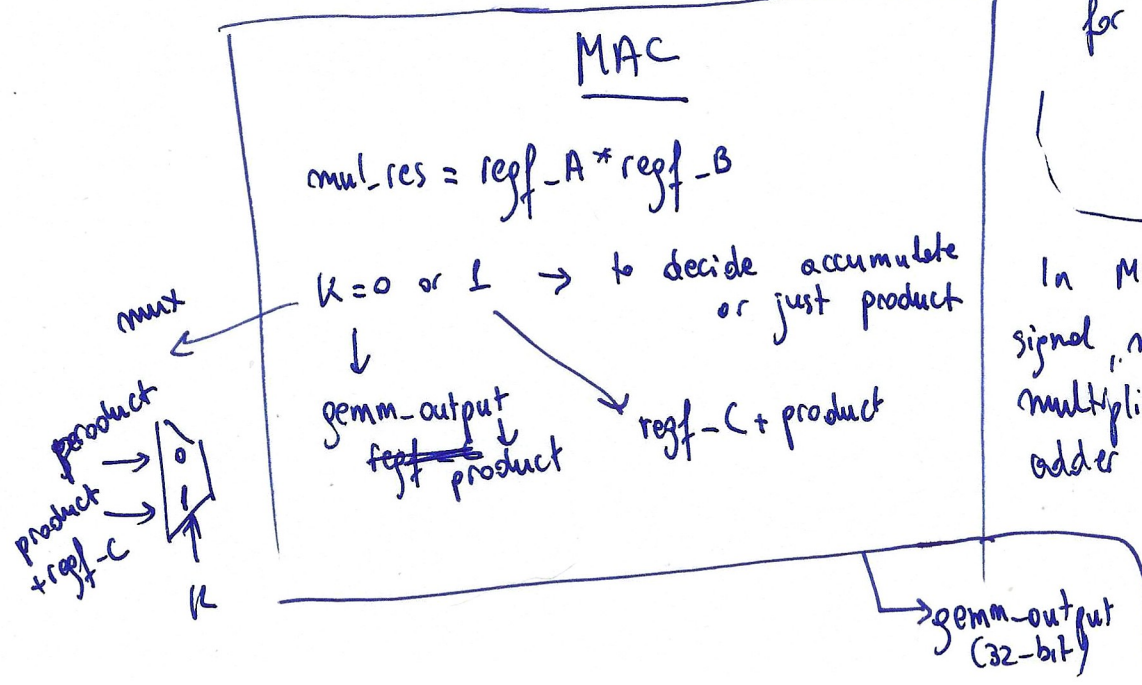
if
 gemm-start 1, active-bank = 0
 gemm-start 2, active-bank = 1

In IDLE, gemm-active-1 & 2 = 0
 ready = 1.

In MAC, ready = 0
 either gemm-active-1 or
 gemm-active-2
 asserted
 for i, j, k same as in PA1,
 k inner loop, j middle, i outer.

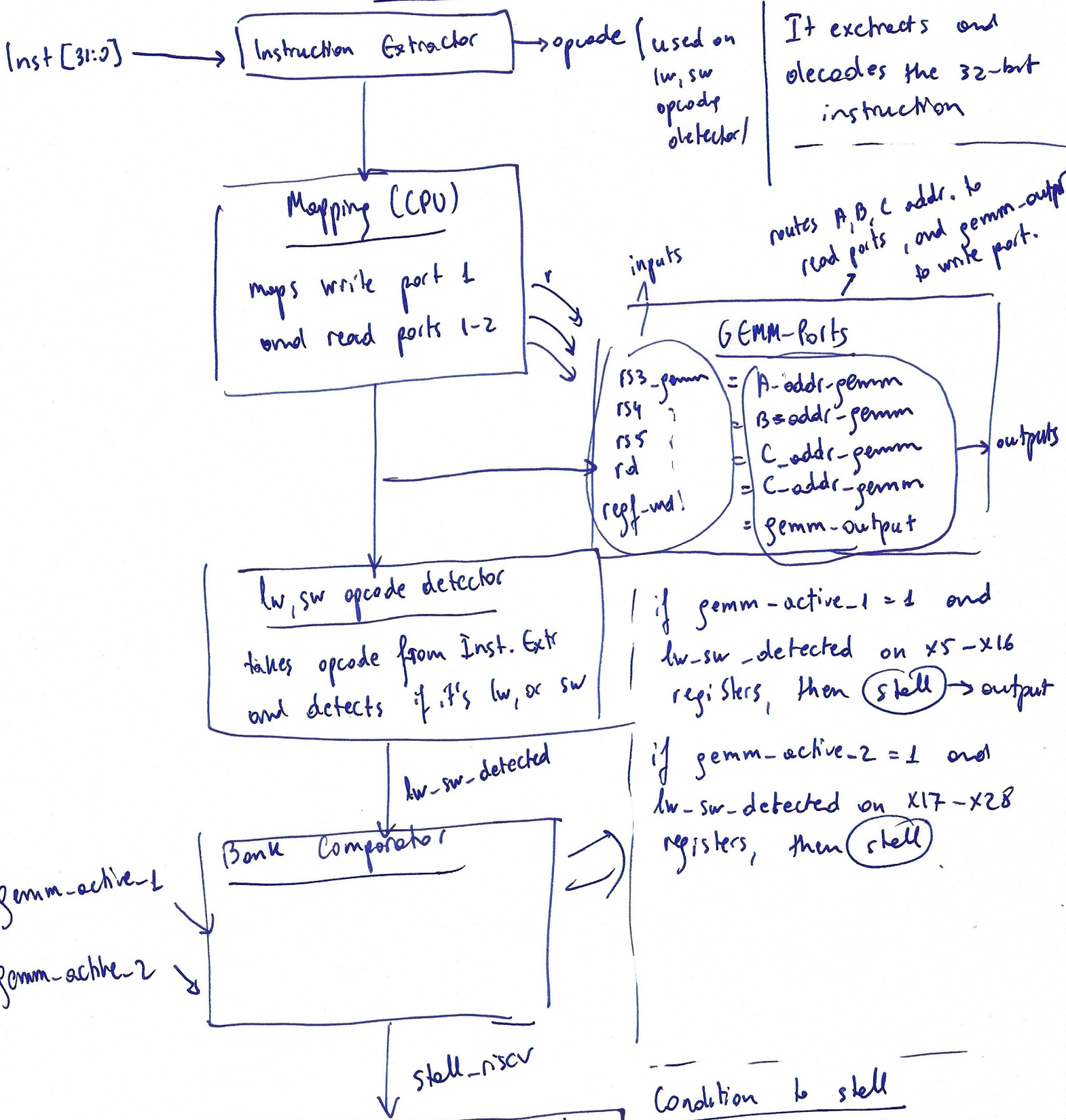
Same as in PA1, it
 generates the address
 of A, B & C.

for gemm-start-1
 ↳ A starts at x5
 B -1- at x9
 C -1- at x13
 for gemm-start-2
 ↳ A starts at x17
 B -1- at x21
 C -1- at x25



In MAC, we have an internal
 signal, mul-res, a 32-bit x 32-bit
 multiplier & a 32-bit full
 adder that calculate product
 & gemm-output. Then
 a k-controlled mux
 decides like in PA1 or PA2

GEMM-RVS



Stall

if stall-riscv signal, then $pc-on=0$ ~~and mem_wrt frozen~~ both inputs & outputs.
 MemWrite Regwrite, riscv-regf-wd, all stall.

Condition to stall

if gemm-active-1 & bank1 conflict. or
 if gemm-active-2 & bank2 - conflict.